DO NOW — RECALL

Client-side versus server-side processing

Name: _____ Class: _____ Date: _____

1. What does an HTML form submit?

2. What does JavaScript textContent do?

3. What does GET generally do with form data?

4. A 2D array has 3 rows and 4 columns. Give total cells and last zero-based indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Name/value fields to a server using a chosen HTTP method.

2. Model answer

Changes text in a DOM node without parsing it as HTML.

3. Model answer

Encodes it as query parameters in the request URL.

4. Model answer

12 cells; last index pair (2,3).

Vocabulary — know these terms

client-side — Processing on user's browser/device.

server-side — Processing on service infrastructure.

validation — Checking data against rules.

authentication — Verifying user identity.

authorisation — Deciding permitted actions.

NEW LEARNING

Read, annotate and explain

Understand

Client-side JavaScript can offer immediate feedback and responsive interfaces, but its validation can be bypassed or altered.

Apply

Server-side code enforces permissions, validates input against business rules and updates authoritative records or databases.

Evaluate

Processing choices trade off device load, server load, confidentiality, latency, accessibility and reliability; many systems use both.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Why must a server check a submitted quantity even if the browser requires 1–5?

Predict first

An attacker can skip the browser form and send a request containing quantity=1000.

Explain

The server must enforce actual constraints; browser checks are useful but only improve convenience.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Compare client-side and server-side processing using form validation. [6]

Question 2

An online shop must deduct stock and charge for an order. Explain which operations cannot rely solely on a browser. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate a hybrid design for an accessible online examination practice system. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A server can trust form data solely because the browser used required.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.